T-Test Homework Report

One-sample, paired-samples, and independent-samples analysis

Course	Statistics for Data Science
Assignment	T-Test analysis
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Executive summary. The required three T-test types were applied to the existing repository dataset. The one-sample and independent tests reject their null hypotheses at alpha = 0.05; the paired comparison does not. All conclusions are descriptive and limited to this synthetic teaching sample.

1. Dataset and method

The analysis uses data/migration_tasks.csv and data/migration_observations.csv. Each migration task has five deterministic synthetic executions. For time-based tests, repeated executions are averaged within ticket_id before testing so that a task, rather than each repeat, is the unit of analysis. The paired test matches agent-specific accuracy by ticket_id.

All tests are two-sided at alpha = 0.05. The one-sample test uses the task-level mean execution time. The paired test uses a paired t-test on agent-A minus agent-B accuracy. The independent comparison uses Welch's t-test because equal variances are not assumed. Effect sizes and 95% confidence intervals are also reported in the generated CSV results.

> Synthetic-data disclosure: execution time, token usage, accuracy, pass rate, and error count are simulated teaching metrics. They are not real AI-agent measurements, Codex benchmarks, production Android measurements, or evidence of general performance.

2. Results

Summary decisions

Test	Comparison	n	t	df	p-value	Decision
One-sample T-test	All migration tasks vs Teaching reference: 75 seconds	36	2.546	35.00	0.01546	Reject H0
Paired-samples T-test	synthetic-agent-A vs synthetic-agent-B	36	0.802	35.00	0.4282	Fail to reject H0
Independent-samples Welch T-test	Hard tasks vs Easy tasks	10	14.002	12.09	7.801e-09	Reject H0

Test	Mean A	Mean B/reference	Difference	95% CI for difference	Effect size
One-sample T-test	88.026	75.000	13.026	[2.639, 23.413]	Cohen's d: 0.424
Paired-samples T-test	0.906	0.901	0.005	[-0.007, 0.016]	Paired Cohen's d_z: 0.134
Independent-samples Welch T-test	130.752	52.506	78.246	[66.080, 90.412]	Hedges' g: 5.672

Interpretation

One-sample: the mean task-level execution time is 88.03 seconds, 13.03 seconds above the 75-second teaching reference. The two-sided p-value is 0.0155, so the null hypothesis is rejected at alpha = 0.05.

Paired: agent A's mean task-level accuracy is 0.906 versus 0.901 for agent B. The mean paired difference is 0.0046 and p = 0.4282, so there is not enough evidence in this synthetic sample to reject a zero mean difference.

Independent: hard tasks average 130.75 seconds versus 52.51 seconds for easy tasks. Welch's p-value is 7.80e-09, so the null hypothesis of equal means is rejected. This reflects the generated difficulty structure, not a production benchmark.

3. Conclusion and reproducibility

The three requested T-test types were completed. In this synthetic dataset, task difficulty is associated with execution time, while the matched agent-accuracy comparison does not show a statistically significant mean difference. These findings are descriptive only and should not be generalized beyond this generated sample.

Reproduksi dari root repository: `python analysis/t_test_analysis.py`. Verifikasi: `python -m unittest discover -s tests -v`.